PUSHFORWARD ALONG A QCQS MORPHISM PRESERVES QUASI-COHERENCE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4994-5037, following the statement at 4986-4992.

Editorial note. The final step 'Thus we conclude that f_*F is a quasi-coherent O_S-module' consumes lemma-kernel-cokernel-quasi-coherent without naming it; the lemma is same-chapter and stated at source line 1636.

Allowed standard prerequisites (editorial).

- Affine characterization of quasi-separated. lemma-characterize-quasi-separated (same chapter, cited at ~5000). Writes each intersection U_i cap U_j as a finite union of affine opens U_ijk.
- Pushforward of tilde along affine morphisms. lemma-widetilde-pullback (same chapter, cited at ~5033). Shows the second and third terms of the exact sequence are quasi-coherent.
- Kernels of maps of quasi-coherent sheaves are quasi-coherent. lemma-kernel-cokernel-quasi-coherent (same chapter, statement at 1636; consumed implicitly at ~5035-5036). Concluding engine for the kernel term.

Source reference. Original lemma 01KO: sources/stacks-schemes/pinned-original-source.tex, source lines 4342–4342. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01I9: sources/stacks-schemes/pinned-original-source.tex, source lines 1459–1459. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. If f is quasi-compact and quasi-separated then f_* transforms quasi-coherent $\mathcal{O}_X$-modules into quasi-coherent $\mathcal{O}_S$-modules.*

Proof. The question is local on S and hence we may assume that S is affine. Because X is quasi-compact we may write $X = \bigcup_{i=1}^{n} U_i$ with each U_i open affine. Because f is quasi-separated we may write $U_i \cap U_j = \bigcup_{k=1}^{n_{ij}} U_{ijk}$ for some affine open U_{ijk}, see Lemma 01KO. Denote $f_i : U_i \to S$ and $f_{ijk} : U_{ijk} \to S$ the restrictions of f.

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

For any open V of S and any sheaf $\mathcal{F}$ on X we have

$$\begin{aligned} f_*\mathcal{F}(V) &= \mathcal{F}(f^{-1}V) \\ &= \mathrm{Ker}\left(\bigoplus\nolimits_i \mathcal{F}(f^{-1}V \cap U_i) \to \bigoplus\nolimits_{i,j,k} \mathcal{F}(f^{-1}V \cap U_{ijk})\right) \\ &= \mathrm{Ker}\left(\bigoplus\nolimits_i f_{i,*}(\mathcal{F}|_{U_i})(V) \to \bigoplus\nolimits_{i,j,k} f_{ijk,*}(\mathcal{F}|_{U_{ijk}})(V)\right) \\ &= \mathrm{Ker}\left(\bigoplus\nolimits_i f_{i,*}(\mathcal{F}|_{U_i}) \to \bigoplus\nolimits_{i,j,k} f_{ijk,*}(\mathcal{F}|_{U_{ijk}})\right)(V) \end{aligned}$$

In other words there is an exact sequence of sheaves

$$0 \to f_*\mathcal{F} \to \bigoplus f_{i,*}\mathcal{F}_i \to \bigoplus f_{ijk,*}\mathcal{F}_{ijk}$$

where $\mathcal{F}_i, \mathcal{F}_{ijk}$ denotes the restriction of $\mathcal{F}$ to the corresponding open. If $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_X$-module then $\mathcal{F}_i$ is a quasi-coherent $\mathcal{O}_{U_i}$-module and $\mathcal{F}_{ijk}$ is a quasi-coherent $\mathcal{O}_{U_{ijk}}$-module. Hence by Lemma 01I9 we see that the second and third term of the exact sequence are quasi-coherent $\mathcal{O}_S$-modules. Thus we conclude that $f_*\mathcal{F}$ is a quasi-coherent $\mathcal{O}_S$-module. $\square$